Nicholas Saroiu

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Education

Honours Bachelor of Science — Computer Science Major, Mathematics Minor — University of Toronto Apr 2026

Relevant Coursework: Neural Networks and Deep Learning; Introduction to Machine Learning; Computer Architecture; Operating Systems; Compilers and Interpreters; Parallel Programming.

Work Experience

Student Researcher — *Lee Language Lab* under supervision of Prof. Annie Lee, Toronto, Ontario Sept 2025 – Present

- Accomplished cross-lingual performance gains in low-resource language models by implementing Punctuation Restoration as an unsupervised learning objective.
- Optimized model effectiveness for multilingual datasets by investigating language-agnostic training behaviors through collaborative experimental design and result analysis.

Jr. Software Engineer — *Dibbly Inc.*, Oakville, Ontario May 2025 – Aug 2025

- Accomplished a more robust and maintainable codebase by introducing automated validation frameworks (Pytest, Jest), reducing regression errors during full-stack AI application deployment.
- Enhanced full-stack application reliability by collaborating within an Agile team of 10 to build responsive features using React.js, Next.js, and Python.

Frontend Developer — *Roca Networks*, Markham, Ontario June 2024 – Oct 2024

- Accomplished the development of an interactive real-time dashboard for residential battery systems, displaying live performance metrics and energy analytics to optimize household energy consumption.
- Enhanced user experience and data clarity by designing responsive React.js components integrated with dynamic Chart.js visualizations.

Projects

Causal Tracing for Transformer Interpretability

- Accomplished the identification of influential internal components in GPT models by applying targeted ablation and restoration to attention heads and activations.
- Reproduced factual association experiments from the NeurIPS 2022 ROME paper "*Locating and Editing Factual Associations in GPT*" by analyzing PyTorch-based interventions to study how models store knowledge.

Models to Predict Student Accuracy

- Accomplished the predictive modeling of student performance by designing, evaluating, and fine-tuning K-Nearest Neighbors, Item-Response Theory, and Autoencoder architectures using PyTorch and scikit-learn.
- Optimized overall predictive performance to achieve a 70% validation accuracy by implementing a robust data preprocessing pipeline, ensemble methods, and rigorous cross-validation techniques.

Patient Portal Backend

- Accomplished a secure, HIPAA-compliant dental patient management system by developing a REST API using Django REST Framework and PostgreSQL.
- Accomplished high-integrity data security by implementing JWT authentication and comprehensive data encryption for sensitive patient records.

TTC Tracker

- Accomplished real-time transit vehicle tracking and arrival predictions by building a full-stack web application for Toronto commuters.
- Developed a responsive React.js frontend with interactive map overlays to visualize live vehicle positions.
- Engineered a Java Spring Boot backend to ingest and process transit GTFS data feeds, ensuring low-latency data synchronization.

Skills

- **Languages:** Python, Java, C, C++, Javascript, Typescript, SQL, MIPS Assembly.
- **Frameworks/Tools:** PyTorch, scikit-learn, NumPy, Git, Docker, SLURM, React.js, Next.js, Docker, Spring Boot, LLVM.
- **Concepts:** Deep Learning and Neural Networks, Transformer Architecture and Interpretability, Model Optimization, Parallel Programming, Performance Profiling, API Design and Integration.